Stochastic Analysis of Successive Interference Cancellation in a Narrow-Beam LEO Uplink

Ilari Angervuori, *Student Member, IEEE* and Risto Wichman, *Senior Member, IEEE*

***Abstract*—We investigate the signal-to-interference ratio (SIR) distributions of user equipments (UEs) and the SIR order statistics at a typical low Earth orbit satellite base station (LEO BS) with a narrow Gaussian antenna in an uplink. We study the SIR distributions of the three strongest UEs under successive interference cancellation (SIC). A Gaussian mixture shadowing model is used. The UEs are located on the Earth according to the Poisson point process (PPP). We demonstrate that with the SIC, each LEO BS can serve multiple UEs in its cell, while achieving a good and stable network throughput.**

I. Introduction and motivation

Successive interference cancellation (SIC) has the potential to improve the performance of multiple access communication systems. The order statistics of the signal-to-interference ratio (SIR) under interference cancellation (and signal combination) have been extensively studied for terrestrial networks [1] using stochastic geometry. In low Earth orbit (LEO) networks, a SIC scheme was studied in [2] using the stochastic geometry method in a non-orthogonal access satellite network. However, a comprehensive stochastic geometry framework for the analysis of the SIR order statistics and the SIC is, to the best of our knowledge, yet to be explored in the LEO networks.

In this paper, we study the SIR under the SIC of user equipments (UEs) at a typical LEO satellite base station (LEO BS) in an urban shadowing environment. We utilize the Gaussian Mixture shadowing model, similar to [3] and [4]. We demonstrate that the link quality can be made more stable with interference cancellation, while maintaining good performance, through increasing the network density while mitigating interference with the SIC, which enables serving multiple UEs at each LEO BS.

In the paper, a highly simplified moment-order-two approximation for log-normal shadowing is proposed and shown to be feasible: a *mixture* exponential distribution with a delta at 0 (*two* degrees of freedom instead of *one*). The exponential fading distribution typically yields the simplest and most insightful results in stochastic geometry analysis, which can be leveraged by this approximation. Furthermore, for the analysis, we propose a planar system, which is feasible for the narrow-beamed LEO BS networks. The analytical results are verified by comparing them to Monte Carlo simulated performance metrics acquired by the realistic system model with a spherical Earth.

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Ilari Angervuori and Risto Wichman are with the Department of Electrical Engineering, Aalto University, Espoo, 02150, Finland. (email: ilari.angervuori@aalto.fi; risto.wichman@aalto.fi).

TABLE I: Glossary of principal symbols and the values used in the simulated numerical results. We denote proportionality "$\propto$" or equality "$=$" to a variable. For a dimensional number, we denote the units "SI" and "[non-SI]."

Symbol	Explanation	Values and units
h	Receiving LEO BS altitudes.	1000 km
α	Power path loss exponent (negligible effect on the SIR).	4
ϵ	The LEO BS elevation angle w.r.t o.	$\{\pi/2, \pi/6\}$ rad $= \{90, 30\}[^\circ]$
λ	The average number of UE transmitters inside a unit area: the density of Φ and the PPP of UEs in the spherical model.	$\{1.44, 0.18\} \times 10^{-3}$/km^2
p_{LoS}	LoS probability; $p_{\text{LoS}} \propto \sin(\epsilon)$.	$\{0.992, 0.493\}$
μ_{LoS}	Mean of the LoS component of the Gaussian mixture shadow fading model.	0 [dB]
μ_{NLoS}	Mean of the NLoS component.	-26 [dB]
σ^2_{LoS}	Variance of the LoS component.	4^2 [dB]
σ^2_{NLoS}	Variance of the NLoS component.	6^2 [dB]
φ_{RX}	Half-width of the -3 dB antenna gain.	$0.028 = 1.6[^\circ]$
κ	Average number of UEs inside a -3 dB footprint; $\kappa = \pi\lambda\left(\varphi_{\text{RX}} h/\sin^2(\epsilon)\right)^2$.	$\{3.53, 7.04\}$
υ	Fraction of effective UEs; $\upsilon \propto \sin(\epsilon)$.	$\{0.85, 0.426\}$
$\kappa\upsilon$	Average number of effective UEs inside a -3 dB footprint (a cell).	3
n	Number of served UEs at each LEO BS.	$n = \kappa\upsilon = 3$
$\tilde{\kappa}$	$\kappa/\log(2)$.	
τ	Minimum SIR for a successful signal decoding.	$0.2 = -7$ [dB]
θ	SIR of the served UE.	$(0.2, 3.16) = (-7, 5)$ [dB]

II. System model

A narrowband LEO uplink is considered with narrow-beamed LEO receivers, and statistics over short use periods are studied. Due to the narrow beam, the Doppler spread can be ignored. The LEO BSs form a homogeneous point process (p.p.) on a plane representing the constellation. Each LEO BS is at altitude $h \in [200, 2000]$ km and elevation angle $\epsilon \in (30, 90)^\circ$. The UEs follow a homogeneous PPP $\Phi \subset \mathbb{R}^2$ of density λ. No fast fading is modeled, and each LEO BS is associated with $n > 0$ served UEs ordered according to their signal powers after log-normal shadowing. Although the noise is not modeled in the work, to ensure that all served UEs have an adequate signal-to-interference-plus-noise ratio (SINR) at the LEO BS [5, Sec. IV-B], n is perceived as the average number of *effective* UEs inside a *cell*, which is defined as the -3 dB footprint of a LEO BS. The term "effective" means here transmitters which are "not entirely" attenuated due to shadowing; we will define this notion precisely in the work.

Figure 1 depicts the planar system model represented on the line. Because of the translation invariance of the PPP, all

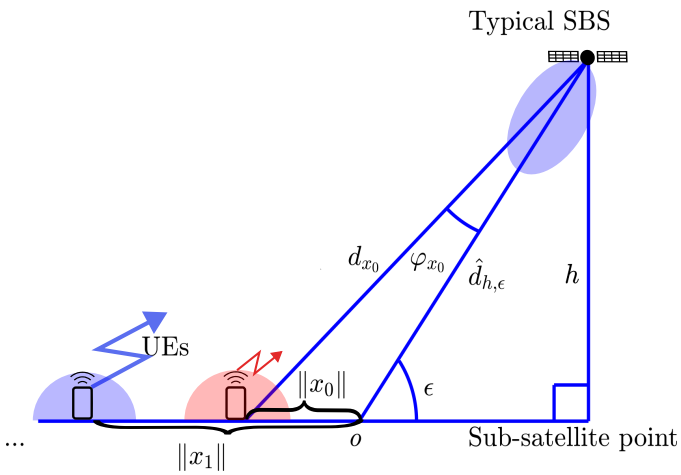

Fig. 1: A sketch of the planar system model. In the figure, the UEs $\{x_i\}$ are projected into the line according to their distance from the origin. The transmitter with the strongest signal is the first-served UE.

locations are statistically equivalent, and we define the origin o to represent the typical LEO BS footprint focus point. We assume a fast-decaying narrow beam for the typical LEO BSs, hence each distance $\hat{d}_{h,\epsilon} \approx d_x$ can be approximated as equal for all (relevant) UEs. Hence, we can normalize $\hat{d}_{h,\epsilon} = 1$ in the analysis. This entails that the power path loss exponent α, as in the spatial path loss $d_x^{-\alpha}$, does not affect the SIR.

The antenna gain at the typical LEO BS is given for the Euclidean distance $\|x\| \in [0, \infty)$ as a Gaussian function

$$G(\|x\|) = 2^{-(D_{h,\epsilon}\|x\|)^2/\varphi_{\text{RX}}^2}, \tag{1}$$

where the angle φ_{RX} denotes the -3 dB antenna gain width. The scaling constant $D_{h,\epsilon} \triangleq \sin^2(\epsilon)/h$ is a first-order coefficient of the Taylor expansion of the angle φ_x w.r.t. the boresight of the typical antenna pattern (see extensive details in [5]).

A. Spherical system model

All analytical results based on the planar model are tested against Monte Carlo simulations based on a system model with a spherical Earth. In the spherical model, all angles and distances are calculated according to spherical geometry. For a more detailed analysis and comparison of the planar model with the spherical model, please refer to [5].

B. Shadowing

1) Gaussian mixture shadowing model (simulations)

Consider a two-tier $\{\text{LoS}, \text{NLoS}\}$ (line-of-sight and non-line-of-sight) Gaussian mixture shadow fading model with the parameters $\mu_{\text{LoS}} = 0$ dB, $\sigma_{\text{LoS}} = 4$ dB, $\mu_{\text{NLoS}} = -26$ dB, and $\sigma_{\text{NLoS}} = 6$ dB, which correspond to an urban environment [6]. Assuming i.i.d. power shadow fading for all UEs, *the typical shadowed transmit power* $H_{\mathcal{MLN}}$ follows a log-normal mixture distribution;

$$\begin{aligned} H_{\mathcal{MLN}} \sim\ & p_{\text{LoS}}\mathcal{LN}(\rho\mu_{\text{LoS}}, (\rho\sigma_{\text{LoS}})^2) \\ & + p_{\text{NLoS}}\mathcal{LN}(\rho\mu_{\text{NLoS}}, (\rho\sigma_{\text{NLoS}})^2), \end{aligned} \tag{2}$$

where $p_{\text{LoS}} = 1 - p_{\text{NLoS}}$ is the LoS probability. Considering a natural base for the log-normal distribution, the constant $\rho \triangleq \log(10)/10$ normalizes the parameters $\mu_{\text{LoS}}, \sigma_{\text{LoS}}, \mu_{\text{NLoS}}$, and σ_{NLoS}, ensuring that the conditioned r.v.'s $10\log_{10}(H_{\mathcal{MLN}}|\text{LoS})$ and $10\log_{10}(H_{\mathcal{MLN}}|\text{NLoS})$ evaluate to r.v.'s following the normal distributions $\mathcal{N}(\mu_{\text{LoS}}, \sigma_{\text{LoS}}^2)$ and $\mathcal{N}(\mu_{\text{NLoS}}, \sigma_{\text{NLoS}}^2)$, respectively.

2) Mixture exponential shadowing distribution (analysis)

As a compromise between analytical tractability and realism, we introduce an exponential distribution with an atomic probability measure at 0, described by the complementary CDF (CCDF)

$$F_{H_{\text{Exp}}}(t) = \upsilon e^{-t}, t > 0. \tag{3}$$

The parameter $0 \leq 1 - \upsilon < 1$ denotes the probability that the shadowed signal is entirely attenuated and takes the value of zero, otherwise, the power follows the exponential distribution.

In the following, the scaling term Υ ensures that the means of the log-normal mixture distribution and the mixture exponential distribution match: $\mathbb{E}(\Upsilon \times H_{\mathcal{MLN}}) = \mathbb{E}(H_{\text{Exp}}) = \upsilon$. By equating the first two moments M1 and M2 of H_{Exp} (l.h.s.) and $\Upsilon \times H_{\mathcal{MLN}}$ (r.h.s.);

$$\begin{cases} \text{M1: } \upsilon = \Upsilon\left(p_{\text{LoS}}e^{\mu_{\text{LoS}}+\sigma_{\text{LoS}}^2/2} + p_{\text{NLoS}}e^{\mu_{\text{NLoS}}+\sigma_{\text{NLoS}}^2/2}\right) \\ \text{M2: } 2\upsilon = \Upsilon^2\left(p_{\text{LoS}}e^{2(\mu_{\text{LoS}}+\sigma_{\text{LoS}}^2)} + p_{\text{NLoS}}e^{2(\mu_{\text{NLoS}}+\sigma_{\text{NLoS}}^2)}\right), \end{cases} \tag{4}$$

we can solve for the parameter υ for the shadowing fading distribution approximation $F_{H_{\text{exp}}}(\cdot)$:

$$\upsilon = \frac{2\left(p_{\text{LoS}}e^{\mu_{\text{LoS}}+\sigma_{\text{LoS}}^2/2} + p_{\text{NLoS}}e^{\mu_{\text{NLoS}}+\sigma_{\text{NLoS}}^2/2}\right)^2}{p_{\text{LoS}}e^{2(\mu_{\text{LoS}}+\sigma_{\text{LoS}}^2)} + p_{\text{NLoS}}e^{2(\mu_{\text{NLoS}}+\sigma_{\text{NLoS}}^2)}}. \tag{5}$$

The parameter $\upsilon = \upsilon(\epsilon)$ varies with the elevation angle through the LoS probability, which influences the shadow fading characteristics. In this sense, υ is the fraction of effective UEs. The parameter Υ holds no significance in an interference-limited scenario, as the equal scaling of all UE powers neutralizes its effect.

The urban scenario LoS probabilities are found in [6, Sec. 6.6.1]. Figure 2 shows the LoS probabilities and $\upsilon = \upsilon(\epsilon)$ w.r.t. the elevation angle. The sine function is also plotted for comparizon.

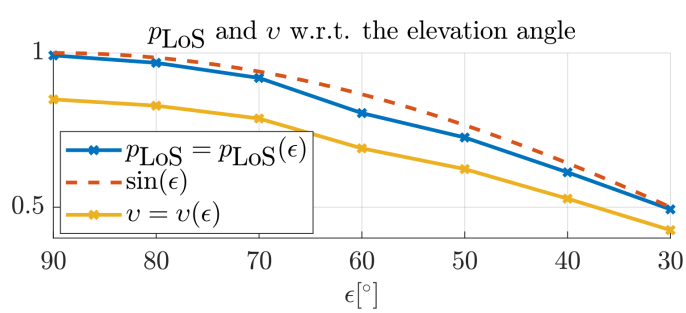

Fig. 2: The LoS probabilities and υ in the urban scenario.

Remark. *The variable $0 < \upsilon \leq 1$ is solvable for almost every shadowing scenario in [6], particularly for the urban scenario.*

III. Analysis

Given i.i.d. shadowing variables $\{H_x\}_{x\in\Phi}$, we define the process of the received signal powers at the typical location, referred to as the gain process (GP), by

$$\mathcal{G} \triangleq \{H_x G(\|x\|) : x \in \Phi\}, \tag{6}$$

where $\|x\|$ is the Euclidean distance from o.

The GP is a *projection process* mapping the points from $\mathbb{R}^2$ into $(0,\infty)$ and, as such, forms a nonhomogeneous PPP [7, Section 4.2.5].

Since the variables $\{H_x\}_{x\in\Phi}$ are i.i.d., we can denote the typical shadowing variable simply as H without the subscript.

Proposition 1 (Density of the GP)**.** *Let $F_H(\cdot)$ be the (possibly degenerate) CCDF of a fading variable H. The density function of $\mathcal{G}$ is given by*

$$\lambda_{\mathcal{G}}(t) = \tilde{\kappa} F_H(t)/t, \quad t \in (0,\infty), \tag{7}$$

where $\tilde{\kappa} = \kappa/\log(2)$ and

$$\kappa \triangleq \pi\lambda \left(\frac{\varphi_{RX} h}{\sin^2(\epsilon)}\right)^2 \tag{8}$$

is approximately the average number of UEs inside a -3 dB footprint (a cell).

Proof. Let $f_H(\cdot)$ be the PDF of H. Denote $G^{-1}(\cdot)$ as the generalized inverse of G, defined as $G^{-1}(y) = \inf\{x : G(x) < y\}$. According to [7, Eq. (4.55)],

$$\begin{aligned}
&\int_t^\infty \lambda_{\mathcal{G}}(y)\,dy = \pi\lambda \mathbb{E}\left[\left(G^{-1}(t/H)\right)^2\right] \\
&= \pi\lambda \int_t^\infty \left(-\frac{\varphi_{\mathrm{RX}}\sqrt{-\log(t/h)}}{D_{h,\epsilon}\sqrt{\log(2)}}\right)^2 f_H(h)\,dh \\
&= -\tilde{\kappa}\int_t^\infty \log(t/h) f_H(h)\,dh \\
&\overset{(a)}{=} -\tilde{\kappa}\left[\log(t/h)F_H(h)|_t^\infty + \int_t^\infty \frac{F_H(h)}{h}\,dh\right].
\end{aligned}$$

In (a), we use integration by parts. The result follows by differentiating with respect to t and applying the negative sign. Note that a necessary condition for this procedure is that $\int_t^\infty \log(t/h) f_H(h)\,dh$ converges for all $t > 0$.

Please refer to [5, Lemma 1] for throughout explanation of the geometric interpretation of κ. □

The total interference, or total received power, is defined as the sum of the GP at the footprint location o of the typical LEO BS:

$$I \triangleq \sum_{x\in\Phi} H_x G(\|x\|) = \sum_{x\in\mathcal{G}} x. \tag{9}$$

The mean and the variance of I are respectively given by

$$\mathbb{E}(I) = \int_0^\infty t\lambda_{\mathcal{G}}(t)dt = \tilde{\kappa}\int_0^\infty F_H(t)dt = \tilde{\kappa}\mathbb{E}(H), \tag{10}$$

$$\begin{aligned}
\mathrm{Var}(I) &= \int_0^\infty t^2\lambda_{\mathcal{G}}(t)dt = \tilde{\kappa}\int_0^\infty tF_H(t)dt \\
&= \tilde{\kappa}\frac{\mathrm{Var}(H) + \mathbb{E}(H)^2}{2} = \tilde{\kappa}\mathbb{E}[H^2]/2.
\end{aligned} \tag{11}$$

Note that matching the first two moments of the fading distributions (2) and (3) is equivalent to matching the mean and the variance of the total interference.

A. Laplace transform of the total received power

With the mixture exponential shadowing $H_{\exp}$,

$$\begin{aligned}
\mathcal{L}_I(s) &\triangleq \mathbb{E}\left(e^{-sI}\right) = \exp\left\{-\int_0^\infty (1-e^{-sr})\lambda_{\mathcal{G}}(r)dr\right\} \\
&= \exp\left\{-\tilde{\kappa}\int_0^\infty (1-e^{-sr})F_{H_{\exp}}(r)/r dr\right\} \\
&= \exp\left\{-\tilde{\kappa}\upsilon\int_0^\infty (1-e^{-sr})e^{-r}/r dr\right\} = (1+s)^{-\tilde{\kappa}\upsilon},
\end{aligned} \tag{12}$$

which is the Laplace transform of the gamma distribution with the shape parameter $\tilde{\kappa}\upsilon$.

B. Order statistics of the STIR and SIR processes

At the typical LEO BS, we denote the signal-to-interference ratio (SIR) process of the UEs as follows:

$$\begin{aligned}
\Psi = \{\mathsf{Z} : \mathsf{Z} \in \Psi\} &\triangleq \left\{\frac{u}{I-u} : u \in \mathcal{G}\right\} \\
&= \left\{\frac{H_x G(D_{h,\epsilon}\|x\|)}{I - H_x G(D_{h,\epsilon}\|x\|)} : x \in \Phi\right\},
\end{aligned} \tag{13}$$

where I is defined in (9). Similarly, the signal-to-total-interference ratio (STIR) process is defined as

$$\Psi' = \{\mathsf{Z}' : \mathsf{Z}' \in \Psi'\} \triangleq \left\{\frac{u}{I} : u \in \mathcal{G}\right\}. \tag{14}$$

We can always recover the process from another:

$$\Psi = \left\{\frac{\mathsf{Z}'}{1-\mathsf{Z}'} : \mathsf{Z}' \in \Psi'\right\}, \quad \Psi' = \left\{\frac{\mathsf{Z}}{1+\mathsf{Z}} : \mathsf{Z} \in \Psi\right\}. \tag{15}$$

Let θ denote the SIR threshold for successful transmission. The event $\Psi \ni \mathsf{Z} > \theta$ is equivalent to $\Psi' \ni \mathsf{Z}' > \theta'$ with $\theta' \triangleq \theta/(1+\theta)$ and $\theta \triangleq \theta'/(1-\theta')$.

We denote $\mathsf{Z}'_{(1)} > \mathsf{Z}'_{(2)} > \mathsf{Z}'_{(3)} \dots$ as the order statistics of the STINR process Ψ', such that $\mathsf{Z}'_{(1)}$ is the largest value in Ψ'. Through the monotonicity of the relations (15), the order statistics of the STIR process are equivalent to the order statistics of the SIR process.

Proposition 2. *The density of the kth factorial moment measure [1, Corollary 10] of the STIR process at the typical LEO BS with a narrow Gaussian antenna beam under the mixture exponential shadowing is given by*

$$\mu'^{(k)}(t'_1,\dots,t'_k) = (\tilde{\kappa}\upsilon)^k \prod_{j=1}^k t'^{-1}_j \left(1 - \sum_{j=1}^k t'_j\right)^{\tilde{\kappa}\upsilon - 1}, \tag{16}$$

whenever $t'_1 > \dots > t'_k$ and $\sum_{i=1}^k t'_i \le 1$, and 0 otherwise.

Proof. By (12), the total interference can be characterized by the gamma process at time $\tilde{\kappa}\upsilon$ [8, Eq. (8)]. Hence, the STIR process Ψ' can be characterized by a Poisson-Dirichlet

distribution $\mathrm{PD}(0, \tilde{\kappa}\upsilon)$ that has the given density [9, Eq. (2.3)]. □

The partial densities can be derived from the density of the kth factorial moment measure as [1, Eq. (62)]

$$\mu'^{(k+i)}_k(z'_1, \ldots, z'_k) = \int_{z'_k}^{1} \cdots \int_{z'_k}^{1} \mu'^{(k+i)}(z'_1, \ldots, z'_k, \zeta'_1, \ldots, \zeta'_i) d\zeta'_1 \ldots d\zeta'_i, \quad (17)$$

the support of the density being in the region $\sum_{j=1}^{n} z'_j + i z'_k \leq 1$.

The joint PDF of the k strongest values of the STIR process $(\mathsf{Z}'_{(1)}, \ldots, \mathsf{Z}'_{(k)})$ is given as a series expansion involving the partial densities [1, Eq. (64)]

$$f'_{(k)}(z'_1, \ldots, z'_k) = \sum_{i=0}^{i_{\max}} \frac{(-1)^i}{i!} \mu'^{(k+i)}_k(z'_1, \ldots, z'_k), \quad (18)$$

for $z'_1 > z'_2 > \cdots > z'_k$ and $f'_{(k)}(z'_1, \ldots, z'_k) = 0$ otherwise. The upper bound for the index $i_{\max} < 1/z'_k - k$ corresponds to the non-zero terms of the series expansion.

Directly from the joint PDF, the k-coverage probability that the first k strongest signals reach the threshold θ is given by

$$\mathcal{P}^{(k)}(\theta) \triangleq \int_{\theta'}^{1} \cdots \int_{\theta'}^{1} f'_{(k)}(z'_1, \ldots, z'_n) dz'_1 \ldots dz'_k, \quad (19)$$

with $\theta' = \theta/(1+\theta)$ and $i_{\max} < 1/\theta' - k$.

The density of the kth factorial moment measure of the SIR process can be extracted from $\mu'^{(k)}$ [7, Corollary 6.1.3]:

$$\mu^{(k)}(z_1, \ldots, z_k) = \prod_{j=1}^{k} \frac{1}{(1+z_j)^2} \mu'^{(k)}\left(\frac{z_1}{1+z_1}, \ldots, \frac{z_k}{1+z_k}\right). \quad (20)$$

C. SIR under interference cancellation

Let $(u_{(1)}, \ldots, u_{(k)}) \subset \mathcal{G}$ represent an ordered set of points in the GP, where $u_{(1)}$ denotes the strongest signal at the typical LEO BS. We denote the SIR of the kth strongest UE as

$$\mathrm{SIR}_{(k)} \triangleq \frac{u_{(k)}}{I - \sum_{j=1}^{k} u_{(j)}}. \quad (21)$$

Let us first study $\mathrm{SIR}_{(1)}$. Combining (16), (18), and (20), we can derive a closed-form for the SIR PDF of the strongest signal in the *simple coverage region* $z \geq 1$: $f_{(1)}(z) = \tilde{\kappa}\upsilon\,(z+1)^{-\tilde{\kappa}\upsilon}/z$ [1]. The first and the second moments of the SIR are bounded by

$$\mathbb{E}(\mathrm{SIR}_{(1)}) \geq \int_1^{\infty} f_{(1)}(z) z dz = \frac{2^{1-\tilde{\kappa}\upsilon}\tilde{\kappa}\upsilon}{\tilde{\kappa}\upsilon - 1}, \quad (22)$$

$$\mathbb{E}(\mathrm{SIR}^2_{(1)}) \geq \int_1^{\infty} f_{(1)}(z) z^2 dz = \frac{2^{1-\tilde{\kappa}\upsilon}(\tilde{\kappa}\upsilon)^2}{(\tilde{\kappa}\upsilon - 1)(\tilde{\kappa}\upsilon - 2)}, \quad (23)$$

respectively. These are divergent for $\tilde{\kappa}\upsilon \leq 1$ and $\tilde{\kappa}\upsilon \leq 2$, respectively: for less than $2\log(2)$ effective UEs inside a cell on average, the first and second moments—hence, also the variance—are infinite (or undefined). Despite the strong average SIR, the infinite variance for $\tilde{\kappa}\upsilon \leq 2$ is not desirable if we want a consistent user experience in link quality.

[1]The SIR has a heavy-tailed distribution, cf. [5, Eq. (32)].

We have the following identity in terms of the STIR process [1, Eq. (69)]:

$$\mathbb{P}(\mathrm{SIR}_{(k)} > \theta) = \mathbb{P}\left(\mathsf{Z}'_{(k)} + \theta' \sum_{j=1}^{k-1} \mathsf{Z}'_{(j)} > \theta'\right). \quad (24)$$

Finally, we consider the SIR under the *perfect* successive signal cancellation (SIC-SIR). [2] Under the SIC, a necessary condition for the successful reception of the kth strongest UE at the typical LEO BS is that the preceding $k-1$ signals are successively decoded and removed from the interference. We denote the minimum SIR threshold for a successful decoding by τ and the corresponding $\tau' = \tau/(1+\tau)$. In this work, we set $\tau = -7$ dB. Formally, the conditions for the SIC-SIR of the kth UE to exceed $\theta \geq \tau$ are

1) $\mathsf{Z}'_{(m)} + \tau' \sum_{j=1}^{m-1} \mathsf{Z}'_j > \tau'$ for all $m \in \{1, \ldots, k-1\}$, and
2) $\mathsf{Z}'_{(k)} + \theta' \sum_{j=1}^{k-1} \mathsf{Z}'_j > \theta'$.

We can use the order statistics theory to study the coverage probability under the SIC.

Proposition 3 (SIC-SIR)**.** *Under the SIC, the coverage probability of the UE with kth strongest signal is given by*

$$\mathcal{P}^{(k)}_{\mathrm{SIC}}(\theta) \triangleq \int_0^1 \cdots \int_0^1 dz'_1 \ldots dz'_k \times f'_{(k)}(z'_1, \ldots, z'_k) \times \mathbb{1}\left(z'_k + \theta' \sum_{j=1}^{k-1} z'_j > \theta'\right) \prod_{m=1}^{k-1} \mathbb{1}\left(z'_m + \tau' \sum_{j=1}^{m-1} z'_j > \tau'\right) \quad (25)$$

with the upper summation limit in $f'_{(k)}(\cdot)$ bounded by $i_{max} < 1/\tau' - 1 = 1/\tau$. $\mathbb{1}(\cdot)$ is the indicator function.

Proof. The expression follows using the joint PDF of the order statistics (18). $i_{\max}$ can be relaxed by the condition $z'_k + \tau' \sum_{j=1}^{k-1} z'_j > \tau'$. By simple algebra, $\sum_{j=1}^{k-1} z_j > 1 - z_k/\tau'$. Recall the condition on the non-zero terms of $\mu'^{(k+i)}$: $\sum_{j=1}^{k} z'_j + i z'_k = \sum_{j=1}^{k-1} z'_j + z'_k + i z'_k \leq 1$. The condition certainly does *not* hold if $1 - z_k/\tau' + z'_k + i z'_k > 1$. We arrive at the inequality $z'_k(-1/\tau' + 1 + i) > 0$. Divide both sides by $z'_k > 0$, and the general upper bound of i follows. □

Considering an adaptive encoding and decoding that adapts according to the SIR of the served UE (after the SIC) at the typical LEO BS, the expected total bandwidth-normalized Shannon throughput [7, Eq. (7.19)] of the n served UEs under the SIC is (see analogous derivation in [5, Eq. (36)])

$$\mathcal{T}_n \triangleq \sum_{k=1}^{n} \mathbb{E}\left(\frac{\log\left(1 + \mathrm{SIR}_{(k)} \prod_{i=1}^{k-1} \mathbb{1}(\mathrm{SIR}_{(i)} > \tau)\right)}{\log(2)}\right) = \sum_{k=1}^{n} \int_{\tau}^{\infty} \frac{\mathcal{P}^{(k)}_{\mathrm{SIC}}(t)}{(t+1)\log(2)} dt \text{ (bit/s/Hz)}. \quad (26)$$

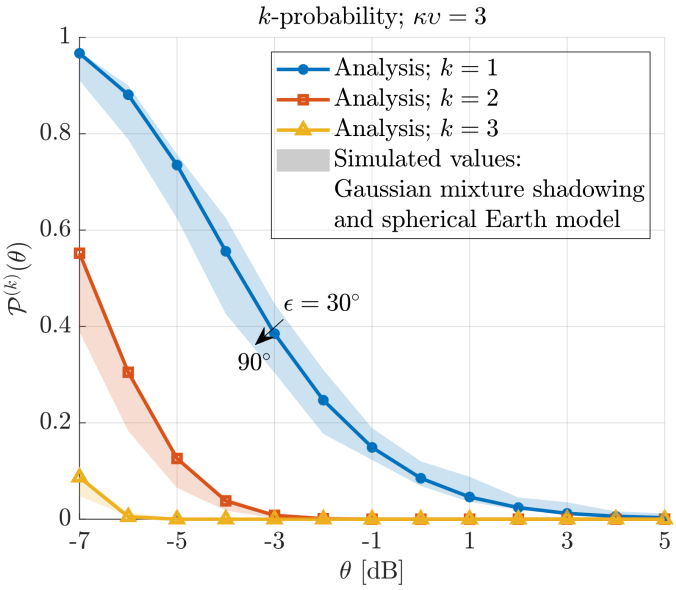

Fig. 3: The SIR k-probabilities of the UEs $x_{(k)} \in \{x_{(1)}, x_{(2)}, x_{(3)}\}$ with $\kappa\upsilon = 3$ (on average, three effective UEs inside each cell).

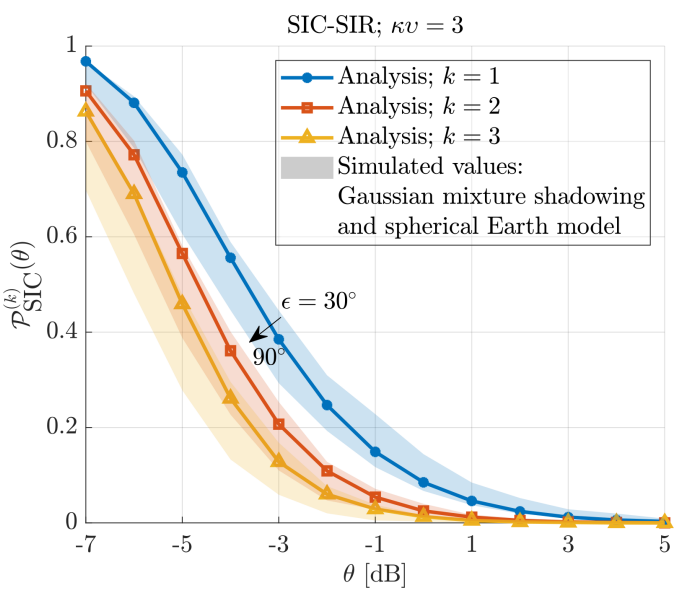

Fig. 4: The SIC-SIR of the UEs $x_{(k)} \in \{x_{(1)}, x_{(2)}, x_{(3)}\}$ with $\kappa\upsilon = 3$ (on average, three effective UEs inside each cell).

IV. Numerical results and conclusions

Figures 3 and 4 depict the k-probabilities (19) and the SIC-SIR (25) for $\kappa\upsilon = \tilde{\kappa}\upsilon \log(2) = n = 3$, respectively. The systems parameters are presented in Table I. In the simulations, λ is determined to match $\kappa\upsilon(\epsilon) = 3$, for each $\epsilon \in \{30, 90\}°$. The simulated coverage probabilities with elevation angles $\epsilon \in (30, 90)°$ fall within the grayed areas in the plots.

The performance of the 2nd and 3rd UEs is significantly enhanced with the SIC-SIR compared to the k-probability, especially for small θ (*i.e.*, low-rate/throughput transmissions), showing a $((0.8620 - 0.0870)/0.0870) \times 100\% \approx 900\%$ improvement for the coverage probability of the third UE with $\theta = -7$ dB.

Note that $\kappa\upsilon = 3$ is in the finite-variance region regarding the SIR distribution of each transmitter (recall, (23)): the link quality is more stable at the LEO BS compared to a smaller UE density scenario, say, $\kappa\upsilon = 1$, when the SIR variance is infinite. This is desirable if we want a consistent user experience. Explicitly, a (loose) upper bound for the variance of the strongest UE in Figures 3 and 4 can be calculated in accordance with (22) and (23): $\mathrm{var}(\mathrm{SIR}_{(1)}) = \mathbb{E}(\mathrm{SIR}^2_{(1)}) - \mathbb{E}(\mathrm{SIR}_{(1)})^2 < 1.3$. In this regard, the SIC-SIR enables more consistent link quality while allowing good performance also for the second and third strongest UEs. Let us study the total throughput next.

Considering the SIR distributions in Figure 4, using the trapezoidal method, we can calculate the throughput (26) as $\mathcal{T}_3 \approx 0.7$ bit/s/Hz. On the other hand, in the infinite-variance SIR region with one served UE in each cell on average ($\kappa\upsilon = n = 1$), we have for the served transmitter $\mathcal{T}_1 \approx 1$ bit/s/Hz: Under the SIC with $\kappa\upsilon = n = 3$, there is a considerable but not overwhelming -0.3 bit/s/Hz tradeoff for link stability (and number of served UEs) as the cost of average network throughput.

This work presents a comprehensive stochastic geometry framework for performance analysis of the typical LEO BS under interference cancellation in the uplink. The model translates to downlink, considering that the cell locations are located according to the PPP on the Earth. Furthermore, the framework extends straightforwardly to signal combination. A clear improvement in the SIR of the second and third strongest UEs was observed under the SIC. The SIC can also be utilized to attain a consistent link quality while maintaining a good average throughput over the LEO BSs.

[2]Imperfect interference cancellation is straightforward to analyze by adding an additional term in $(0, 1)$ [7, Eq. (6.21)] to (24).

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